the center of the loop, indicating that the field present near the center of the coil is almost uniform (Figure 3.34).

The strength of the magnetic field is increased if either the current in the coil or the number of turns or both are increased. The polarity (north pole or south pole) depends on the direction of current in the loop.

3.7.3 Right hand thumb rule

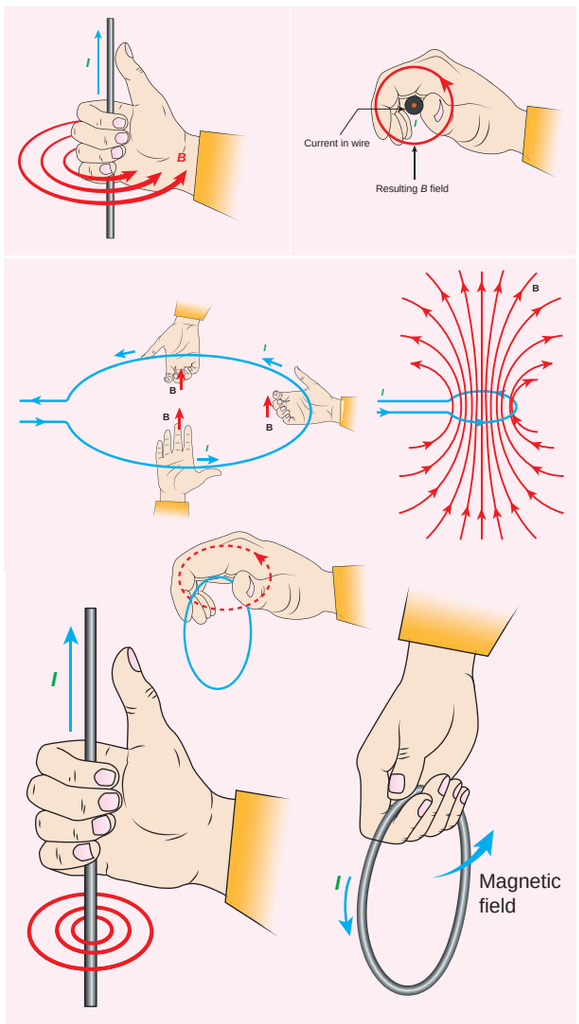


Figure 3.35 Right hand rule – straight conductor and circular loop

The right hand rule is a mnemonic to find the direction of magnetic field when the direction of current in a conductor is known.

If we hold the current carrying conductor in our right hand such that the thumb points in the direction of current flow, then the fingers encircling the wire points in the direction of the magnetic field lines produced.

The Figure 3.35 shows the right hand rule for current carrying straight conductor and circular coil.

Note

Mnemonic means that it is a special word or a collection of words used to help a person to remember something.

3.7.4 Maxwell's right hand cork screw rule

This rule is used to determine the direction of the magnetic field. If we rotate a right-handed screw using a screw driver, then the direction of current is same as the direction in which screw advances and the direction of rotation of the screw gives the direction of the magnetic field. (Figure 3.36)

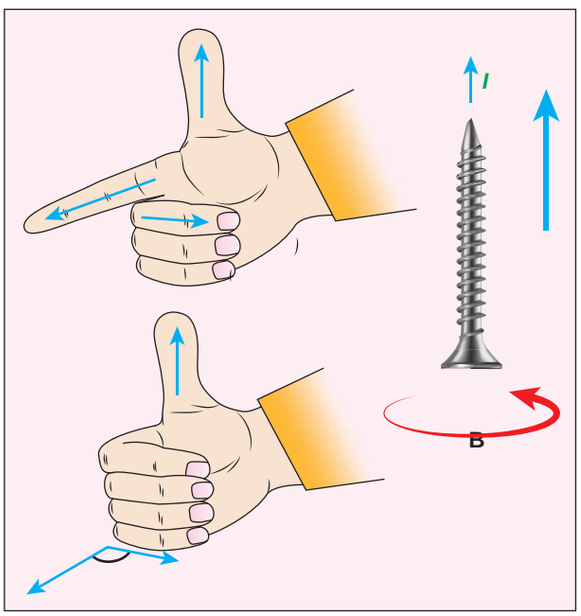


Figure 3.36 Maxwell's right hand cork screw rule

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